

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12407971
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Olson; Kyle et al.

Ear-wearable electronic device including multi-function removal handle

Abstract

An ear-wearable electronic device comprises a shell and a faceplate connected to the shell. Electronic circuitry and a power source are respectively disposed in the shell. A removal handle includes a proximal end connected to the faceplate. First and second electrical conductors extend along the removal handle and comprise first and second proximal ends coupled to the electronic circuitry. An electrical contact module is disposed at a distal end of the removal handle. The electrical contact module comprises a substrate. A first electrical contact is mounted on, or supported by, the substrate and coupled to a first distal end of the first electrical conductor. A second electrical contact is mounted on, or supported by, the substrate and coupled to a second distal end of the second electrical conductor.

Inventors: Olson; Kyle (St. Louis Park, MN), Higgins; Sidney A. (Maple Grove, MN), Stewart; Jay (Eden Prairie, MN), Yang; Fue (Minneapolis, MN), Murray; Casey (Waconia, MN)

Applicant: STARKEY LABORATORIES, INC. (Eden Prairie, MN)

Family ID: 83506461

Assignee: Starkey Laboratories, Inc. (Eden Prairie, MN)

Appl. No.: 17/948464

Filed: September 20, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230098807 A1	Mar. 30, 2023

Related U.S. Application Data

Publication Classification

Int. Cl.: **H04R1/10** (20060101); **H04R25/00** (20060101)

U.S. Cl.:

CPC **H04R1/1058** (20130101); **H04R1/1025** (20130101); **H04R1/1041** (20130101);
H04R1/105 (20130101); **H04R25/609** (20190501); **H04R25/65** (20130101);

Field of Classification Search

CPC: H04R (1/1058); H04R (25/609); H04R (1/1025); H04R (1/1041); H04R (1/105); H04R (25/65)

USPC: 381/374

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7446720	12/2007	Victorian et al.	N/A	N/A
10034107	12/2017	Nielsen	N/A	N/A
10687156	12/2019	Polinske et al.	N/A	N/A
2009/0208046	12/2008	Schmidt	N/A	N/A
2012/0170781	12/2011	Klemenz	381/323	H02J 7/0042
2014/0254844	12/2013	Shennib	N/A	N/A
2015/0146901	12/2014	Richardson et al.	N/A	N/A
2019/0268706	12/2018	Solum	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3180927	12/2016	EP	N/A

OTHER PUBLICATIONS

European Patent Application No. 22198114, filed Sep. 27, 2022; European Search Report issued Mar. 3, 2023; 10 pages. cited by applicant

Primary Examiner: Nguyen; Sean H

Attorney, Agent or Firm: Mueting Raasch Group

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/250,637, filed Sep. 30, 2021, and U.S. Provisional Application

TECHNICAL FIELD

(1) This application relates generally to removal handles for ear-wearable devices, such devices including ear-wearable electronic devices, hearing aid, hearables, earbuds, personal amplification devices, and physiologic/biometric monitoring devices.

SUMMARY

(2) Some embodiments are directed to an ear-wearable electronic device comprising a shell and a faceplate connected to the shell. Electronic circuitry and a power source are respectively disposed in the shell. A removal handle includes a proximal end connected to the faceplate. First and second electrical conductors extend along the removal handle and comprise first and second proximal ends coupled to the electronic circuitry. An electrical contact module is disposed at a distal end of the removal handle. The electrical contact module comprises a substrate. A first electrical contact is mounted on, or supported by, the substrate and coupled to a first distal end of the first electrical conductor. A second electrical contact is mounted on, or supported by, the substrate and coupled to a second distal end of the second electrical conductor.

(3) Some embodiments are directed to an ear-wearable electronic device comprising a shell and a faceplate connected to the shell. Electronic circuitry and a rechargeable power source are respectively disposed in the shell. A removal handle is connected to the faceplate. First and second electrical conductors extend along the removal handle and comprise first and second proximal ends coupled to the electronic circuitry. An electrical contact module is disposed at a distal end of the removal handle. The electrical contact module comprises a substrate. A first electrical contact is mounted on, or supported by, the substrate and coupled to a first distal end of the first electrical conductor. A second electrical contact is mounted on, or supported by, the substrate and coupled to a second distal end of the second electrical conductor. The first and second electrical contacts are configured as touch sensor contacts and charge contacts for charging the rechargeable power source.

(4) Some embodiments are directed to a method implemented using an ear-wearable electronic device comprising a shell and a faceplate connected to the shell. The method comprises, with the device deployed in the wearer's ear, activating a switch of an electrical contact module disposed at a distal end of a removal handle connected to the faceplate in response to contact between at least one of the wearer's fingers and the electrical contact module. The method also comprises, with the device removed from the wearer's ear, recharging a rechargeable power source of the device via the electrical contact module.

(5) The above summary is not intended to describe each disclosed embodiment or every implementation of the present disclosure. The figures and the detailed description below more particularly exemplify illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Throughout the specification reference is made to the appended drawings wherein:

(2) FIG. 1 shows a representative ear-wearable electronic device which includes a multi-function removal handle in accordance with any of the embodiments disclosed herein;

(3) FIG. 2 shows a multi-function removal handle in accordance with any of the embodiments disclosed herein;

(4) FIG. 3A shows various components of a multi-function removal handle in accordance with any of the embodiments disclosed herein;

(5) FIG. 3B shows various components of a multi-function removal handle in accordance with any

of the embodiments disclosed herein;

(6) FIG. 4A shows various components of a multi-function removal handle in accordance with any of the embodiments disclosed herein;

(7) FIG. 4B shows various components of a multi-function removal handle in accordance with any of the embodiments disclosed herein;

(8) FIG. 5 is a block diagram of an ear-wearable electronic device which includes a multi-function removal handle configured to serve as one or both of a user-actuatable control comprising a touch sensor and charge contacts in accordance with any of the embodiments disclosed herein;

(9) FIG. 6 shows a touch sensor of an ear-wearable electronic device and an output signal generated by the touch sensor, the touch sensor comprising first and second charge contacts of a multi-function removal handle in accordance with any of the embodiments disclosed herein;

(10) FIG. 7 illustrates a touch detection circuit integral or coupled to a user-actuatable control of an ear-wearable electronic device in accordance with any of the embodiments disclosed herein;

(11) FIG. 8 is a schematic of rectifier circuitry for electrically connecting charge contacts of a multi-function removal handle and charging contacts of a charger unit in accordance with any of the embodiments disclosed herein; and

(12) FIG. 9 is a schematic of rectifier circuitry for electrically connecting charge contacts of a multi-function removal handle and charging contacts of a charger unit in accordance with any of the embodiments disclosed herein.

(13) The figures are not necessarily to scale. Like numbers used in the figures refer to like components. However, it will be understood that the use of a number to refer to a component in a given figure is not intended to limit the component in another figure labeled with the same number.

DETAILED DESCRIPTION

(14) Removal handles for hearing aids and other in-canal electronic devices provide a gripping structure which allows easy extraction of the device from a wearer's ear. Removal handles are also called pull-cables, pull-cords, and pull-members. Embodiments of the disclosure are directed to removal handles with enhanced functionality.

(15) In some implementations, a multi-function removal handle comprises a user-actuatable touch switch. In other implementations, the multi-function removal handle comprises charge contacts configured to couple to charging contacts of a charging unit. In further implementations, the multi-function removal handle comprises a user-actuatable touch switch and charge contacts configured to couple to charging contacts of a charging unit. In some implementations, the charge contacts are self-orienting, such that the charge contacts automatically assume a preferred orientation (e.g., electrical polarity) when placed in the charging unit. In other implementations, the positioning of the charge contacts can be arbitrary, due to rectifying circuitry coupled to the charge contacts.

(16) In further implementations, the multi-function removal handle includes a metal core tube through which electrical conductors pass. The metal core tube provides enhanced strength for the removal handle, which can be subjected to significant tensile and torsional forces when pulled on by the device wearer. In some implementations, the multi-function removal handle includes a metal core tube which serves as a radiating element of an antenna of the ear-wearable electronic device. In other implementations, the multi-function removal handle includes an outer sleeve of conductive material that serves as a radiating element of an antenna of the ear-wearable electronic device. The outer conductive sleeve can be covered by a protective coating, such as a coating to prevent oxidation.

(17) The multi-function removal handle can incorporate one or more of these and other advantageous features disclosed herein. Embodiments of the disclosure provide for custom rechargeable hearing devices and other ear-wearable electronic devices (e.g., standard non-rechargeable hearing devices) which provide for enhanced removal handle functionality in a space-saving manner.

(18) Many custom in-canal devices are simply too space limited or reside too deeply in the ear

canal for a pushbutton to be effective. Such devices include in-the-ear (ITE), in-the-canal (ITC), completely-in-the-canal (CIC), and invisible-in-canal (IIC) type hearing devices. In cases where a pushbutton does fit and is accessible, such custom in-canal devices often press against the tragus causing soreness and long-term wear that the user must expect to enjoy the added functionality. (19) Typically, charging contacts reside on the faceplate of custom in-canal devices which is presently a space constraint component that further limits fit rate and comfort shaping by their presence. The faceplate location also limits design options and increases size of the charger while also making the creation of a reliable Travel Charger all but impossible. The addition of wires and Kevlar reinforcement in a removal handle weaken the cross-sectional shear strength of the removal handle by the necessary thinning of the wall or co-extrusion polymer to maintain the desired sectional diameter. Embodiments of ear-wearable electronic devices which incorporate a multi-function removal handle address these and other shortcomings in conventional removal handle technology.

(20) Embodiments of the disclosure are defined in the claims. However, below there is provided a non-exhaustive listing of non-limiting examples. Any one or more of the features of these examples may be combined with any one or more features of another example, embodiment, or aspect described herein.

(21) Example Ex1. An ear-wearable electronic device comprises a shell and a faceplate connected to the shell, electronic circuitry and a power source respectively disposed in the shell, a removal handle having a proximal end connected to the faceplate, first and second electrical conductors extending along the removal handle and comprising first and second proximal ends coupled to the electronic circuitry, and an electrical contact module disposed at a distal end of the removal handle. The electrical contact module comprises a substrate, a first electrical contact mounted on, or supported by, the substrate and coupled to a first distal end of the first electrical conductor, and a second electrical contact mounted on, or supported by, the substrate and coupled to a second distal end of the second electrical conductor.

(22) Example Ex2. The device according to Ex1, wherein the electrical contact module defines a user-actuable switch.

(23) Example Ex3. The device according to Ex1 or Ex2, wherein the switch defines a single function switch.

(24) Example Ex4. The device according to Ex1 or Ex2, wherein the switch defines a multiple function switch.

(25) Example Ex5. The device according to one or more of Ex1 to Ex4, wherein the power source comprises a rechargeable power source, and the electrical contact module defines a user-actuable switch and the first and second electrical contacts define first and second charge contacts configured to facilitate charging of the rechargeable power source.

(26) Example Ex6. The device according to one or more of Ex1 to Ex4, wherein the power source comprises a rechargeable power source, and the first and second electrical contacts define first and second charge contacts configured to facilitate charging of the rechargeable power source.

(27) Example Ex7. The device according to one or more of Ex1 to Ex6, comprising a magnet disposed on or in the substrate, wherein the magnet is configured to magnetically interact with a magnet of a charging unit so as to self-orient the electrical contact module on the charging unit.

(28) Example Ex8. The device according to one or more of Ex1 to Ex7, wherein the substrate is disposed between the first and second electrical contacts and serves as a dielectric positioned therebetween, and the substrate is a rigid structure or a flexible structure.

(29) Example Ex9. The device according to Ex8, wherein the substrate comprises a printed circuit board (PCB).

(30) Example Ex10. The device according to one or more of Ex1 to Ex9, wherein the substrate comprises an over-molded polymeric element.

(31) Example Ex11. The device according to one or more of Ex1 to Ex10, wherein the removal

handle comprises a metallic tube through which the first and second electrical conductors extend, and the metallic tube is configured as a radiating element of an antenna of the ear-wearable electronic device.

(32) Example Ex12. The device according to one or more of Ex1 to Ex10, wherein the removal handle comprises a tube through which the first and second electrical conductors extend, and a metallic sleeve is disposed over the tube and is configured as a radiating element of an antenna of the ear-wearable electronic device.

(33) Example Ex13. The device according to one or more of Ex1 to Ex12, comprising rectifying circuitry coupled to the first and second electrical contacts.

(34) Example Ex14. The device according to one or more of Ex1 to Ex13, wherein the device defines an ITE, ITC, CIC or IIC type hearing device.

(35) Example Ex15. An ear-wearable electronic device comprises a shell and a faceplate connected to the shell, electronic circuitry and a rechargeable power source respectively disposed in the shell, a removal handle connected to the faceplate, first and second electrical conductors extending along the removal handle and comprising first and second proximal ends coupled to the electronic circuitry, and an electrical contact module disposed at a distal end of the removal handle. The electrical contact module comprises a substrate, a first electrical contact mounted on, or supported by, the substrate and coupled to a first distal end of the first electrical conductor, a second electrical contact mounted on, or supported by, the substrate and coupled to a second distal end of the second electrical conductor, and the first and second electrical contacts configured as touch sensor contacts and charge contacts for charging the rechargeable power source.

(36) Example Ex16. The device according to Ex15, wherein the shell has a uniquely-shaped outer surface that corresponds uniquely to an ear geometry of a wearer of the device.

(37) Example Ex17. The device according to Ex15 or Ex16, wherein the electrical contact module is configured as a single function touch sensor.

(38) Example Ex18. The device according to Ex15 or Ex16, wherein the electrical contact module is configured as a multiple function touch sensor.

(39) Example Ex19. The device according to one or more of Ex15 to Ex18, wherein the device defines an ITE, ITC, CIC or IIC type hearing device.

(40) Example Ex20. A method implemented using an ear-wearable electronic device comprising a shell and a faceplate connected to the shell comprises, with the device deployed in the wearer's ear, activating a switch of an electrical contact module disposed at a distal end of a removal handle connected to the faceplate in response to contact between at least one of the wearer's fingers and the electrical contact module, and with the device removed from the wearer's ear, recharging a rechargeable power source of the device via the electrical contact module.

(41) Example Ex21. The method according to Ex20, comprising magnetically orienting the electrical contact module on a charging unit in a preferred charging orientation.

(42) Example Ex22. The method according to Ex20 or Ex21, wherein the switch defines a single function switch.

(43) Example Ex23. The method according to Ex20 or Ex21, wherein the switch defines a multiple function switch.

(44) Example Ex24. The method according to one or more of Ex20 to Ex23, wherein the device defines an ITE, ITC, CIC or IIC type hearing device.

(45) FIG. 1 shows a representative ear-wearable electronic device 5 comprising a shell 11 having a uniquely-shaped outer surface that corresponds uniquely to an ear geometry of a wearer of the device 5. In some implementations, the shell 11 can have a standard shape useful by a population of wearers. Although not shown in FIG. 1 (but see FIG. 5), electronic circuitry and a power source are respectively disposed in the shell 10. In some embodiments, the power source comprises a rechargeable power source, such as a lithium-ion battery. In other embodiments, the power source comprises a non-rechargeable power source, such as a zinc-air battery (e.g., a 312 zinc-air battery).

A faceplate **12** is disposed at a proximal end of the shell **10**. A multi-function removal handle **14** extends from the faceplate **12** in an outer ear direction.

(46) The multi-function removal handle **14** includes an elongated tube **16** and an electrical contact module **20** disposed at a distal end **19** of the tube **16**. In some implementations, the electrical contact module **20** is configured as a user-actuatable touch sensor or switch. In other implementations, the electrical contact module **20** includes charge contacts configured to facilitate charging of a rechargeable power source of the device **5** when the charge contacts electrically couple to an external charging unit. In further implementations, the electrical contact module **20** is configured as a user-actuatable touch sensor or switch and, in addition, provides charge contacts for electrically coupling to a charging unit.

(47) FIGS. **2**, **3A-3B**, and **4A-4B** illustrate various components of the multi-function removal handle **14** shown in FIG. **1** in accordance with any of the embodiments disclosed herein. As shown, the electrical contact module **20** is disposed at a distal end **19** of the tube **16** of the multi-function removal handle **14**.

(48) According to some embodiments, and as shown in FIG. **3B**, the tube **16** includes a metal core tube **21** covered by a sheath of polymeric material **23** over-molded onto the core tube **21**. In some implementations, the tube **16** is a metal weave core tube. Enhanced sheer and tensile strength are derived from this metal lattice. The outer jacket of polymeric material includes blunts formed on each end to anchor the multi-function removal handle **14** in the faceplate **12** and mechanically lock onto a substrate (e.g., PCB **22**) or other structure of the electrical contact module **20**. In some implementations, a polymeric tube can be used instead of the metal core tube **21**.

(49) In some embodiments, the metal core tube **21** is configured to serve as a radiating element of an antenna of the ear-wearable electronic device **5**. In the case of a metal weave core tube, such as that shown in FIG. **3B**, the tightness of the weave (or spiral in some configurations) enables radiofrequency (RF) radiation to bridge these small gaps, which results in an antenna structure having an effective size close to the outer diameter of the removal handle **14** (in contrast to the size of a small wire contained inside a polymeric tube as in the case of a conventional implementation). Antenna performance is unaffected by processes (e.g., electrical activity) inside the metal core tube **21** as radiation propagation is directed outwardly from the outer surface of the metal core tube **21**. The antenna comprising the metal core tube **21** can be coupled to a communication device (e.g., communication device **172** shown in FIG. **5**) of the ear-wearable electronic device **5** via a wire arrangement, an inductive coupling arrangement, or a magnetic coupling arrangement.

(50) In some embodiments, and with continued reference to FIG. **3B**, item **23** can be implemented as a conductive (e.g., metal) sleeve which extends along all or a portion of the removal handle **14**. A dielectric material can be disposed between the metal core tube **21** and the conductive sleeve **23**. In other embodiments, item **21** can be implemented as a polymeric tube and item **23** can be implemented as the conductive sleeve **23** which extends along all or a portion of the removal handle **14**. A protective coating can be disposed on the conductive sleeve **23** to prevent oxidation. The conductive sleeve **23** is configured to serve as a radiating element of an antenna of the ear-wearable electronic device **5**. The antenna comprising the conductive sleeve **23** can be coupled to a communication device of the ear-wearable electronic device via a wire arrangement, an inductive coupling arrangement, or a magnetic coupling arrangement.

(51) It is noted that the embodiments incorporating the conductive sleeve **23** discussed above can be implemented in a receiver-in-canal (MC) cable. The MC cable can have a construction similar to that described above with reference to FIG. **3B**. For example, item **21** can be implemented as a polymeric tube with wires extending within the tube between a behind-ear hearing device unit and a receiver disposed at a distal end of the polymeric tube. A conductive sleeve **23** extends along at least a portion of the RIC cable. The conductive sleeve **23** is configured to serve as a radiating element of an antenna of the RIC device. The antenna comprising the conductive sleeve **23** can be coupled to a communication device of the MC device via a wire arrangement, an inductive

coupling arrangement, or a magnetic coupling arrangement.

(52) With reference to FIG. 3A, the electrical contact module **20** includes a substrate in the form of PCB **22**, a first electrical contact **24**, and a second electrical contact **26**. The first electrical contact **24** is mounted on a first surface of the PCB **22** and coupled to a first distal end of the first electrical conductor **25**. The second electrical contact **26** is mounted on a second surface of the PCB **22** and coupled to a second distal end of the second electrical conductor **27**.

(53) As shown in FIG. 4, first and second electrical conductors **25**, **27** extend along the interior of the tube **16**. The first and second electrical conductors **25**, **27** comprise first and second proximal ends which are coupled to electronic circuitry disposed within the shell **10**. The first and second electrical conductors **25**, **27** extend up the tube **16** and to the PCB **22** where they are soldered to electrical pads **30**, **32**. The pads **30**, **31** are connected to traces which respectively terminate on the first and second electrical contacts **24**, **26**.

(54) As can be seen in FIGS. 2 and 3A, the PCB **22** is disposed between the first and second electrical contacts **24**, **26** and serves as a dielectric positioned therebetween. In some implementations, the PCB **22** is a rigid structure (e.g., FR-4). In other implementations, the PCB **22** is a flexible structure (e.g., a flexible polymeric (e.g., polyimide) printed circuit board). In some embodiments, the electrical contact module **20** includes a third electrical contact mounted on a third surface of the PCB **22**. In such embodiments, the PCB **22** can have at least three surfaces (e.g., planar surfaces) each of which supports one of the three electrical contacts. The PCB **22** is disposed between the first, second, and third electrical contacts and serves as a dielectric positioned therebetween.

(55) According to some embodiments, the electrical contact module **20** includes a magnet **34** mounted on a specified surface of the PCB **22**. The magnet **34** is configured to magnetically interact with a magnet of an external charging unit so as to self-orient the electrical contact module **20** when placed on the charging unit. For example, when the wearer places the ear-wearable electronic device **5** in a charging unit, such as a travel charge unit, the electrical contact module can automatically flip to the correct magnetic polarity (orientation) which ensures that the correct electrical polarity is achieved for charging. It is noted that the charging unit can have recessed portions on the base and lid configured to receive the device **5** and the multi-function removal handle **14**. When the lid is closed, a gentle (continuous) squeeze force can be applied to the first and second electrical contacts **24**, **26** which is interpreted by the device **5** as a charging command. Removal of the device **5** from the charger terminates the squeeze force, thereby terminating the charging procedure.

(56) FIGS. 4A and 4B illustrate a multi-function removal handle in accordance with any of the embodiments disclosed herein. The multi-function removal handle **50** includes an electrical contact module **50** which comprises a polymeric substrate **52**, a first electrical contact **54**, and a second electrical contact **56**. The polymeric substrate **52** can be an over-molded polymeric element. The polymeric substrate can include a void **53** (e.g., a cylindrical void). The first electrical contact **54** is mounted on, or supported by, a first surface of the substrate **52**. The first electrical contact **54** is coupled to a first distal end of a first electrical conductor **55** via electrical pad **60**. The second electrical contact **56** is mounted on, or supported by, a second surface of the substrate **52**. The second electrical contact **56** is coupled to a first distal end of a second electrical conductor **57** via electrical pad **62**.

(57) As shown in FIG. 4B, first and second electrical conductors **55**, **57** extend along the interior of the tube **66**. The first and second electrical conductors **55**, **57** comprise first and second proximal ends which are coupled to electronic circuitry disposed within the shell **10**. The first and second electrical conductors **55**, **57** extend up the tube **66** and to the substrate **52** where they are soldered to electrical pads **60**, **62**. The pads **60**, **62** can be connected to traces which respectively terminate on the first and second electrical contacts **54**, **56**.

(58) As is shown in FIG. 4A, the substrate **52** is disposed between the first and second electrical

contacts **54**, **56** and serves as a dielectric positioned therebetween. In some implementations, the substrate **52** is a rigid over-molded polymeric structure. In other implementations, the substrate **52** is a flexible over-molded polymeric structure. In some embodiments, the electrical contact module **50** includes a third electrical contact mounted on, or supported by, a third surface of the substrate **52**. In such embodiments, the substrate **52** can have at least three surfaces (e.g., planar surface) each of which supports one of the three electrical contacts. The substrate **52** is disposed between the first, second, and third electrical contacts and serves as a dielectric positioned therebetween.

(59) According to some embodiments, the electrical contact module **50** includes a magnet **64** disposed on or in the substrate **52**. As shown, the magnet **64** is disposed within a void **53** of the substrate **52**. Although shown to be cylindrical, the magnet **64** and void **53** can have any shape (e.g., a polygonal shape). As in other embodiments, and as previously discussed, the magnet **64** is configured to magnetically interact with a magnet of an external charging unit so as to self-orient the electrical contact module **50** when placed on the charging unit.

(60) In some implementations, the electrical contact module **20**, **50** serves as a single function touch switch which execute a specified function (e.g., on/off, volume change cycling between low and high, cycling through memory settings). In other implementations, the electrical contact module **20**, **50** serves as a multiple function touch switch which executes a multiplicity of specified functions (see examples below).

(61) In accordance with any of the embodiments disclosed herein, and with reference to FIGS. **5** and **6**, the ear-wearable electronic device **100** can be configured as a hearing device or a hearable which includes an audio processing facility **170**. The audio processing facility **170** can include audio signal processing circuitry, a speaker, receiver or other acoustic transducer, and optionally one or more microphones. In accordance with any of the embodiments disclosed herein, the device **100** can be implemented as a physiologic (e.g., biometric) monitoring device and include a sensor facility **134**. When implemented for physiologic monitoring (e.g., via a heart rate sensor, blood oxygen saturation sensor, respiration sensor, fall detector), the device **100** can include or exclude (e.g., be devoid of) the audio processing facility **170**.

(62) The device **100** can also incorporate a communication device **172** configured to effect communications between two of the devices **100** and/or with an external electronic device, system, and/or the cloud. The communication device **172** can include one or both of an RF transceiver/antenna and/or an NFMI/NFC transceiver/antenna. For example, the communication device **172** can include one or more radios that conform to an IEEE 802.11 (e.g., WiFi®) or Bluetooth® (e.g., BLE, Bluetooth®) specification, for example. It is understood that the device **100** can employ other radios, such as a 900 MHz radio. In addition, or alternatively, the device **100** can include a near-field magnetic induction (NFMI) sensor (e.g., an NFMI transceiver coupled to a magnetic antenna) for effecting short-range communications (e.g., ear-to-ear communications, ear-to-kiosk communications). The communication device **172** is coupled to an antenna of the device **100**, such as a metallic tube antenna or a metallic sleeve antenna as disclosed herein.

(63) The shell **102** of the device **100** is configured to contain or support a number of components including some or all of those shown in FIG. **5**. The device **100** also includes a multi-function removal handle **140** connected to, and extending from, the shell **102**. The multi-function removal handle **140** shown in FIG. **5** is preferably the same as the multi-function removal handle **14** shown in FIGS. **1-4**.

(64) The multi-function removal handle **140** includes a tube **152** having a proximal end connected to the faceplate **150** and a distal end configured to support an electrical contact module **138**. In addition to serving as a pull cord, the multi-function removal handle **140** can include charge contacts **146** according to various embodiments. In other embodiments, in addition to serving as a pull cord, the multi-function removal handle **140** can include a user-actuatable control **132**, such as a touch sensor. In further embodiments, in addition to serving as a pull cord, the multi-function removal handle **140** can include charge contacts **146** and the user-actuatable control **132**.

(65) According to embodiments in which the electrical contact module **138** includes charge contacts **146**, the charge contacts **146** are electrically coupled to charging circuitry **145** disposed in the shell **102**. The charging circuitry **145** is coupled to a rechargeable power source **144**, such as a lithium-ion battery. In some implementations, the charge contacts **146** are coupled to rectifier circuitry **155** which, in turn, is coupled to the charging circuitry **145**. The rectifier circuitry **155** can be configured according to the schematics shown in FIGS. **8** and **9**. The rectifier circuitry **155** provides for any orientation of the charge contacts **146** (e.g., either-way OK circuitry) when electrically coupled to corresponding electrical contacts of an external charging unit.

(66) In some implementations (e.g., those in which rectifier circuitry may not be included), a magnet **142** is disposed in the electrical contact module **138**. The magnet **142** is configured to magnetically interact with a magnet of a charging unit so as to self-orient the electrical contact module **138** relative to electrical contacts of the charging unit. When the wearer places the ear-wearable electronic device **100** in a charging unit, such as a travel charge unit for example, the electrical contact module **138** can automatically flip to the correct magnetic polarity (orientation) which ensures that the correct electrical polarity is achieved for charging the rechargeable power source **144**.

(67) A controller **120** is operatively coupled to the charging circuitry **145** and other components of the device **100**. The charge contacts **146** typically comprise at least one positive contact and at least one negative or ground contact exposed on an exterior surface of the electrical contact module **138** (e.g., see electrical contacts **24** and **26** shown in FIG. **3**). When recharging the rechargeable power source **144**, the device **100** is typically placed in a charging unit comprising positive and negative charge contacts which electrically couple to corresponding positive and negative charge contacts **146** of the electrical contact module **138**. The charging circuitry **145** of the device **100** cooperates with charging circuitry of the charging unit to charge the rechargeable power source **144**. The charging circuitry **145** of the device **100** can be configured to cooperate with the charging unit in accordance with the representative examples disclosed in commonly owned, co-pending U.S. Published Patent Application No. 2019/0386,498, which is incorporated herein by reference in its entirety.

(68) The electrical contact module **138** of device **100** can also include a user-actuable control **132** (e.g., a touch sensor or switch). The user-actuable control **132** comprises or is coupled to the charge contacts **146**. It is noted that, other than the charge contacts and electrical conductors coupled to same, some or all of the circuitry of the user-actuable control **132** can be disposed in the shell **102**.

(69) The electrical contact module **138** can be implemented as a sensor comprising the charge contacts **146** and configured to sense for a change in one or any combination of impedance, conductance, resistance, and electrodermal activity. More particularly, the electrical contact module **138** can be implemented as a sensor that uses the charge contacts **146** to sense for a change in one or any combination of impedance, conductance, resistance, and electrodermal activity due to the presence of the wearer's finger or fingers at the electrical contact module **138** (referred to herein as a touch input).

(70) In accordance with any of the embodiments disclosed herein, and as shown in FIG. **6**, the electrical contact module **138** is configured to sense contact between the wearer's finger or fingers and the charge contacts **146** as a touch input. The user-actuable control **132** generates a touch signal **151** in response to the touch input sensed by the electrical contact module **138**. The representative touch signal **151** shown in FIG. **2** is a pulse **153** (e.g., a voltage or current pulse) generated by signal processing circuitry of the user-actuable control **132**. The user-actuable control **132** communicates the pulse **153** or an output signal corresponding to the pulse **153** to the controller **120**. The controller **120** is configured to initiate a device function or operation or a plurality of device functions or operations in response to the touch input pulse **153** or a sequence of touch input pulses **153**.

(71) Although a single pulse **153** is shown in the representative example of FIG. **6**, it is understood that multiple pulses **153** can be generated by the user-actuatable control **132** and operated on by the controller **120** to perform a number of different operations. Multiple pulses **153** can be generated in response to multiple touches to the electrical contact module **138**. The pulse width of the pulse or pulses **153** can be varied so as to correspond to individual, or a sequence of, short and long touches. The number, duration, and/or time separation of wearer touches to the electrical contact module **138** and corresponding pulses **153** can correspond to a number of different device commands or wearer inputs.

(72) For example, a single touch or pinch to the electrical contact module **138** followed by a long pause (e.g., a single tap input) can be interpreted by the controller **120** as corresponding to a volume up command. Two touches or pinches to the electrical contact module **138** separated in time by a short pause (e.g., a double tap input) can be interpreted by the controller **120** as corresponding to a volume down command. A single long duration touch or pinch to the electrical contact module **138** can be interpreted by the controller **120** as corresponding to a communication device (e.g., telecoil, RF transceiver) activation command. Two long duration touches or pinches to the electrical contact module **138** can be interpreted by the controller **120** as corresponding to a communication device deactivation command.

(73) Multiple commands for a given device function or operation can also be input to the device **100** via the electrical contact module **138**. For example, a short duration touch followed by a long duration touch or pinch applied to the electrical contact module **138** can initiate a hearing aid memory selection routine. Once initiated, application of individual short duration touches or pinches to the electrical contact module **138** can allow the wearer to step/cycle through a number of different hearing aid memories, each of which activates different hearing aid settings (e.g., speech enhancement settings, noise reduction settings, ambient environment classification settings). It is understood that these and other examples of hearing device functions described herein represent non-exhaustive, non-limiting examples of device commands that can be initiated by a wearer using a user-actuatable control facility disclosed herein.

(74) FIG. **7** illustrates a touch detection circuit **180** integral or coupled to a user-actuatable control **132** of an ear-wearable electronic device **100** in accordance with any of the embodiments disclosed herein. Typically, some or all of the components of the touch detection circuit **180** (other than the electrical contacts and corresponding electrical conductors) are disposed in the shell **102** of the device **100**, given the limited space of the multi-function removal handle **140**. The touch detection circuit **180** can be implemented as an ASIC, such as a 4-pin ASIC. The components used to construct the touch detection circuit **180** are widely available and inexpensive, and the circuit **180** can be less expensive than a single mechanical switch.

(75) The touch detection circuit **180** is electrically coupled to positive and negative contacts **147**, **148** of the electrical contact module **138**. The touch detection circuit **180** includes a first section **181** configured to electrically sense contact between a wearer's finger(s) **143** and the positive and negative contacts **147**, **148** of the electrical contact module **138**. The first section **181** includes a first field effect transistor (FET) **Q1**, such as a PMOS FET, with a gate, *g*, coupled to a charging circuit line **185** connected to positive contact **147**. A source, *s*, of **Q1** is coupled to a voltage source having a voltage of *V_{dd}* (a battery voltage). A drain, *d*, of **Q1** is coupled to an output **182** over which a Touch Signal #1 can be communicated to the user-actuatable control **132** and/or controller **120** of the device **100**. The first section **181** also includes a resistor **R1** having a relatively high resistance, such as 10 MΩ, coupled between the charging circuit line **185** and the source voltage *V_{dd}* line. It is noted that a pulldown resistor **R2** is typically connected to the output **182** for sensing Touch Signal #1, but the resistor **R2** need not be part of the touch detection circuit **180**.

(76) A second section **183** of the touch detection circuit **180** can be included in some implementations or excluded in others. The second section **183** includes a thermistor **141a** coupled to the negative or ground contact **148**. The thermistor **141a** can be a component of the electrical

contact module **138** and configured to thermally sense contact or near-contact between a wearer's finger **143** and the electrical contact module **138**. A third section **184** of the touch detection circuit **180** includes a second field effect transistor (FET) **Q2**, such as a PMOS FET, with a gate, *g*, coupled to voltage source *Vdd*, a source, *s*, coupled to the charging circuit line **185**, and a drain, *d*, coupled to a charging circuit (e.g., a power management IC of charging circuitry **145**) of the device **100**.

(77) An ESD (electrostatic discharge) or TVS (transient voltage suppression) diode **149** can be connected between the positive and negative contacts **147**, **148** to prevent damage to, or unintentional activation of, the circuit **180** in response to static discharge from a user's finger **143**. The ESD/TVS diode **149** preferably has a very low leakage current to avoid artificially pulling the gate of **Q1** low.

(78) When a wearer touches the electrical contact module **138**, the wearer's finger acts as a high impedance (e.g., about 10 k Ω to 1M Ω) applied across the charge contacts **147**, **148**. The application of this high impedance across the charge contacts **147**, **148** pulls the gate, *g*, of **Q1** low, thereby turning on **Q1**. Charge contact **147** is pulled up to the source voltage *Vdd*. Turning on **Q1** results in a voltage signal equal to *Vdd* to be communicated to the output **182** as Touch Signal #1.

(79) The second FET, **Q2**, is configured to prevent a touch event at the electrical contacts **147**, **148** from being incorrectly interpreted as a charging event. A charging event can be initiated in response to application of a charging voltage across charge contacts **147**, **148** that exceeds the source voltage, *Vdd*, by at least the threshold voltage, *Vgs.sub.th*, of **Q2**. For example, the charging voltage (e.g., ~5 V) that turns on **Q2** for charging is greater than *Vdd* (e.g., 4.0 V)+*Vgs.sub.th* (e.g., 0.5 V). In other words, the source voltage, *Vdd*, must be less than the charge voltage, *Vchg*, minus the threshold voltage, *Vgs.sub.th*, of **Q2** (e.g., $Vdd < Vchg - Vgs.sub.th$).

(80) In some embodiments, the electrical contact module **138** can include a temperature sensor which can enhance the accuracy of touch detections (e.g., reduce or eliminate false detections as a separate, corroborating touch sensor). A thermistor **141a** can be coupled to the negative charge contact **148**, which is the low thermal impedance output side of the circuit **180**. When the wearer's finger **143** makes contact or near-contact with the electrical contact module **138**, the thermistor **141a** immediately changes in temperature and quickly approaches the temperature of the user's finger **143**. This rapid change in temperature results in a corresponding change in an analog signal which is communicated from an output **190** of the thermistor **141a** (corresponding to Touch Signal #2) to a temperature ADC input of the user-actuable control **132** and/or controller **120** of the device **100**. The controller **120** can use Touch Signal #1 (electrical touch signal) and Touch Signal #2 (thermal touch signal) for implementing any of the touch detection techniques disclosed herein. It is noted that, unlike capacitive touch solutions, water and hair will not affect the thermistor **141a** (or other temperature sensor) in the same way the finger will.

(81) To further combat false positives and make the electrical contact module **138** more robust, an IMU (disposed in the shell) or a capacitive touch sensor can be used as an auxiliary sensor. In the case of the IMU, a double tap to the faceplate of the shell could be required to activate the components of the electrical contact module **138** (e.g., thermistor **141a** or other temperature touch sensor). In the case of a capacitive touch sensor **134d**, for example, a touch would be recognized if both the capacitance changed as well as the temperature. In some configurations, the charge contacts **146** can be used as electrodes coupled to typical skin on/off circuits or to measure the galvanic skin resistance of the wearer's finger if it can be assured that the touch is applied between the charge contacts **147**, **148**.

(82) FIG. 8 shows a two-contact rectifier circuit in which charge contacts **1** and **2** are contacts to the outside world. As can be seen in FIG. 8, the charger pins CHG+ and CHG- retain their polarity as shown irrespective of the orientation of charge contacts **1** and **2** in the charging unit. FIG. 9 shows a three-contact rectifier circuit in which charge contacts **1**, **2**, and **3** are contacts to the outside world. As can be seen in FIG. 9, the charger pins CHG+ and CHG- retain their polarity as

shown irrespective of the orientation of charge contacts 1, 2, and 3 in the charging unit.

(83) Although reference is made herein to the accompanying set of drawings that form part of this disclosure, one of at least ordinary skill in the art will appreciate that various adaptations and modifications of the embodiments described herein are within, or do not depart from, the scope of this disclosure. For example, aspects of the embodiments described herein may be combined in a variety of ways with each other. Therefore, it is to be understood that, within the scope of the appended claims, the claimed invention may be practiced other than as explicitly described herein.

(84) All references and publications cited herein are expressly incorporated herein by reference in their entirety into this disclosure, except to the extent they may directly contradict this disclosure. Unless otherwise indicated, all numbers expressing feature sizes, amounts, and physical properties used in the specification and claims may be understood as being modified either by the term “exactly” or “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the foregoing specification and attached claims are approximations that can vary depending upon the desired properties sought to be obtained by those skilled in the art utilizing the teachings disclosed herein or, for example, within typical ranges of experimental error.

(85) The recitation of numerical ranges by endpoints includes all numbers subsumed within that range (e.g. 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5) and any range within that range. Herein, the terms “up to” or “no greater than” a number (e.g., up to 50) includes the number (e.g., 50), and the term “no less than” a number (e.g., no less than 5) includes the number (e.g., 5).

(86) The terms “coupled” or “connected” refer to elements being attached to each other either directly (in direct contact with each other) or indirectly (having one or more elements between and attaching the two elements). Either term may be modified by “operatively” and “operably,” which may be used interchangeably, to describe that the coupling or connection is configured to allow the components to interact to carry out at least some functionality (for example, a radio chip may be operably coupled to an antenna element to provide a radio frequency electric signal for wireless communication).

(87) Terms related to orientation, such as “top,” “bottom,” “side,” and “end,” are used to describe relative positions of components and are not meant to limit the orientation of the embodiments contemplated. For example, an embodiment described as having a “top” and “bottom” also encompasses embodiments thereof rotated in various directions unless the content clearly dictates otherwise.

(88) Reference to “one embodiment,” “an embodiment,” “certain embodiments,” or “some embodiments,” etc., means that a particular feature, configuration, composition, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. Thus, the appearances of such phrases in various places throughout are not necessarily referring to the same embodiment of the disclosure. Furthermore, the particular features, configurations, compositions, or characteristics may be combined in any suitable manner in one or more embodiments.

(89) The words “preferred” and “preferably” refer to embodiments of the disclosure that may afford certain benefits, under certain circumstances. However, other embodiments may also be preferred, under the same or other circumstances. Furthermore, the recitation of one or more preferred embodiments does not imply that other embodiments are not useful and is not intended to exclude other embodiments from the scope of the disclosure.

(90) As used in this specification and the appended claims, the singular forms “a,” “an,” and “the” encompass embodiments having plural referents, unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.

(91) As used herein, “have,” “having,” “include,” “including,” “comprise,” “comprising” or the like are used in their open-ended sense, and generally mean “including, but not limited to.” It will be understood that “consisting essentially of” “consisting of,” and the like are subsumed in

“comprising,” and the like. The term “and/or” means one or all of the listed elements or a combination of at least two of the listed elements.

(92) The phrases “at least one of,” “comprises at least one of,” and “one or more of” followed by a list refers to any one of the items in the list and any combination of two or more items in the list.

Claims

1. An ear-wearable electronic device, comprising: a shell and a faceplate connected to the shell; electronic circuitry and a rechargeable power source respectively disposed in the shell; a removal handle having a proximal end connected to the faceplate, the removal handle comprising an elongated tubular structure; first and second electrical conductors extending along the removal handle and comprising first and second proximal ends coupled to the electronic circuitry; and an electrical contact module disposed at a terminus of a distal end of the elongated tubular structure, the electrical contact module comprising: a substrate; a first electrical contact mounted on, or supported by, the substrate and coupled to a first distal end of the first electrical conductor; and a second electrical contact mounted on, or supported by, the substrate and coupled to a second distal end of the second electrical conductor; the first and second electrical contacts defining first and second charge contacts configured to facilitate charging of the rechargeable power source.
2. The device according to claim 1, wherein the electrical contact module defines a user-actuatable switch.
3. The device according to claim 2, wherein the switch defines a single function switch.
4. The device according to claim 2, wherein the switch defines a multiple function switch.
5. The device according to claim 1, wherein: the electrical contact module defines a user-actuatable switch, the first and second electrical contacts configured as touch sensor contacts of the user-actuatable switch.
6. The device according to claim 1, comprising a magnet disposed on or in the substrate, wherein the magnet is configured to magnetically interact with a magnet of a charging unit so as to self-orient the electrical contact module on the charging unit.
7. The device according to claim 1, wherein: the substrate is disposed between the first and second electrical contacts and serves as a dielectric positioned therebetween; and the substrate is a rigid structure or a flexible structure.
8. The device according to claim 7, wherein the substrate comprises a printed circuit board (PCB).
9. The device according to claim 1, wherein the substrate comprises an over-molded polymeric element.
10. The device according to claim 1, wherein: the elongated tubular structure of the removal handle comprises a metallic tube through which the first and second electrical conductors extend; and the metallic tube is configured as a radiating element of an antenna of the ear-wearable electronic device.
11. The device according to claim 1, wherein the elongated tubular structure of the removal handle comprises a tube through which the first and second electrical conductors extend; and a metallic sleeve is disposed over the tube and is configured as a radiating element of an antenna of the ear-wearable electronic device.
12. The device according to claim 1, comprising rectifying circuitry coupled to the first and second electrical contacts.
13. The device according to claim 1, wherein the device defines an ITE, ITC, CIC or IIC type hearing device.
14. An ear-wearable electronic device, comprising: a shell and a faceplate connected to the shell; electronic circuitry and a rechargeable power source respectively disposed in the shell; a removal handle connected to the faceplate, the removal handle comprising an elongated tubular structure; first and second electrical conductors extending along the removal handle and comprising first and

second proximal ends coupled to the electronic circuitry; and an electrical contact module disposed at a terminus of a distal end of the elongated tubular structure, the electrical contact module comprising: a substrate; a first electrical contact mounted on, or supported by, the substrate and coupled to a first distal end of the first electrical conductor; a second electrical contact mounted on, or supported by, the substrate and coupled to a second distal end of the second electrical conductor; and the first and second electrical contacts configured as touch sensor contacts and charge contacts for charging the rechargeable power source.

15. The device according to claim 14, wherein the shell has a uniquely-shaped outer surface that corresponds uniquely to an ear geometry of a wearer of the device.

16. The device according to claim 14, wherein the electrical contact module is configured as a single function touch sensor.

17. The device according to claim 14, wherein the electrical contact module is configured as a multiple function touch sensor.

18. The device according to claim 14, wherein the device defines an ITE, ITC, CIC or IIC type hearing device.

19. A method implemented using an ear-wearable electronic device comprising a shell and a faceplate connected to the shell, the method comprising: with the device deployed in the wearer's ear, activating a switch of an electrical contact module disposed at a terminus of a distal end of an elongated, tubular removal handle connected to the faceplate in response to contact between at least one of the wearer's fingers and the electrical contact module; and with the device removed from the wearer's ear, recharging a rechargeable power source of the device via the electrical contact module.

20. The method according to claim 19, comprising magnetically orienting the electrical contact module on a charging unit in a preferred charging orientation.

21. The method according to claim 19, wherein the switch defines a single function switch.

22. The method according to claim 19, wherein the switch defines a multiple function switch.

23. The method according to claim 19, wherein the device defines an ITE, ITC, CIC or IIC type hearing device.
